Document Identifier: **PMC8085971**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not required.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
DEGs between a comparison of randomly picked two transcriptomes among the seven were characterized by DEseq2 twofold cut-off and adjusted P-value (q-value) <0.01 and followed by ANOVA analysis to identify genes that were statistically differentially expressed among the seven growth conditions (degrees_freedom1 = 6 and degrees_freedom2 = 14, P-value < 0.05).
Blinding
not detected.
Power Analysis
not detected.
Replication
Two to six independent repeats were performed for each target gene.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
After blocking with 5 non-fat milk in TBST for 1 h, the membranes for Cra and RpoB were, respectively, incubated with polyclonal anti-DDDDK (FLAG) tag (Abcam, Cambridge, MA, USA) and monoclonal anti-RpoB antibody (BioLegend, San Diego, CA, USA) for 1.5 h, washed 3 times with TBST, subsequently blotting with the goat anti-rabbit IgG and goat anti-mouse IgG HRP-conjugated secondary antibody (Abmart, Berkeley Heights, NJ, USA) for 1.5 h, and washed three times with TBST.	anti-DDDDK		
	FLAG		
	anti-RpoB		
Recombinant DNA			
For over-expression of Cra and Fur, full-length cra and fur genes from NCM3722 were amplified by PCR and both cloned into pET24a via Nde IXho I, generating pET24a-cra and pET24a-fur, respectively.	pET24a		Suggestion: RRID:Addgene_109422(link)
	pET24a-cra		
	pET24a-fur		
PCR products obtained by using primers listed in Table S1 were sub-cloned into T-vector pMD19 simple (Takara), and single colonies were picked and subjected to sequencing.	pMD19		
Software and Algorithms			
DEGs between a comparison of randomly picked two transcriptomes among the seven were characterized by DEseq2 twofold cut-off and adjusted P-value (q-value) <0.01 and followed by ANOVA analysis to identify genes that were statistically differentially expressed among the seven growth conditions (degrees_freedom1 = 6 and degrees_freedom2 = 14, P-value < 0.05).	DEseq2		Suggestion: DESeq2 (RRID:SCR_015687)(link)
Heatmap visualization and KEGG pathway-enriched analysis Both tools for heatmap visualization and KEGG pathway-enriched analysis of the transcriptome data were conducted with OmicShare online platform (www.omicshare.comtools) (GENE DENOVO) .	KEGG		

Consensus of Crp and Cra binding sequences were generated by WebLogo 3 (Schneider and Stephens, 1990).	WebLogo		Suggestion: WEBLOGO (RRID:SCR_010236)(link)
Signals corresponding to each protein were quantified with ImageJ (NIH) and presented as a relative amount to the level of that in Glc.	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
FbP analysis was performed using 10 ml of sample loaded into a Nexera UPLC (SHIMADZU) with an ACQUITY UPLC C18 column (100 x 2.1 mm, 1.7 mm, Waters) and underwent gradient elution procedures then the sample was ionized by ESI and detected by QE high-resolution mass spectrometer (Thermo) with Xcalibur 4.0 workstation in the negative ion mode, scanning precursor ion mz 338.99 for FbP.	Xcalibur		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
DEGs between a comparison of randomly picked two transcriptomes among the seven were characterized by DEseq2 twofold cut-off and adjusted P-value (q-value) <0.01 and followed by ANOVA analysis to identify genes that were statistically differentially expressed among the seven growth conditions (degrees_freedom1 = 6 and degrees_freedom2 = 14, P-value < 0.05).	ANOVA
Pearson correlation coefficient R between the expression of all the DEGs identified at the seven carbon substrates and the corresponding GR were then calculated.	Pearson correlation

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq